

ASYMPTOTIC DISTRIBUTION THEORY IN THE ANALYSIS OF COVARIANCE STRUCTURES (A UNIFIED APPROACH)

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Summary: In this paper we present a unified approach to the asymptotic distribution theory of covariance structures. Our approach is based on the differential calculus of min-max functions, which is developed in the first sections of the paper. The general theory is demonstrated by means of generalized least squares estimators. New and well-known results are obtained in the general framework. In particular, the asymptotic behaviour of the estimators under an alternative hypothesis and the problem of nonidentifiability are discussed. Finally, potential possibilities of the theory are illustrated by the example of minimum trace factor analysis.

1. INTRODUCTION

In covariance structure analysis it is hypothesized that the $p \times p$ population covariance matrix Σ with elements σ_{ij} is generated by q true though unknown parameters that can be assembled in the $q \times 1$ vector $\underline{\theta}$. Thus, under a given model, each element σ_{ij} of the covariance matrix is a function $\sigma_{ij} = \sigma_{ij}(\underline{\theta})$ of $\underline{\theta} \in \Omega$, where Ω is a specified parameter space. It may be said that $\Sigma = \Sigma(\underline{\theta})$ describes (in matrix notation) the particular structure under investigation that relates the parameter vector $\underline{\theta}$ to Σ . For example in the well-known factor analysis model one assumes that

$$(1.1) \quad \Sigma = \Lambda\Lambda' + \Psi$$

where Λ denotes a $p \times k$ matrix of factor loadings and Ψ is a $p \times p$ diagonal matrix. Here the structural model is given by (1.1) with the parameter vector $\underline{\theta}$ consisting of elements of Λ and Ψ .

Another example is the linear structure (cf., Anderson (1973))

$$(1.2) \quad \Sigma = \theta_1 \Sigma_1 + \dots + \theta_q \Sigma_q$$

where $\Sigma_1, \dots, \Sigma_q$ are known symmetric matrices and $\underline{\theta} = (\theta_1, \dots, \theta_q)'$ is the parameter vector. For instance, in the simple case $q = 1$ and $\Sigma_1 = I_p$ (identity matrix) it is hypothesized that all variances are equal and all covariances are zero.

Among the statistical problems of covariance structure analysis are the estimation of parameters of the model, establishing properties of the estimators and evaluation of goodness of fit. For a given sample covariance matrix S based on $N = n + 1$ observations one may try to estimate $\underline{\theta}$ by solving the system of $p(p+1)/2$ (generally nonlinear) equations

$$(1.4) \quad s_{ij} = \sigma_{ij}(\underline{\theta}), \quad i < j = 1, \dots, p.$$

However, if Ω is an open subset of R^q , the functions $\sigma_{ij}(\underline{\theta})$, $i, j = 1, \dots, p$ are continuously differentiable and $q < p(p+1)/2$ then equations (1.4) cannot be solved *exactly* for *almost every* symmetric matrix S . This result follows immediately from the very general Theorem of Sard (e.g., Ortega and Rheinboldt (1970), p.131). If the set Ω is defined by r equality constraints $\Omega = \{\underline{\theta} \in R^q : g_i(\underline{\theta}) = 0, i=1, \dots, r\}$, where $g_i(\underline{\theta})$, $i=1, \dots, r$ are continuously differentiable functions, then the (local) dimension of Ω is $q-r$ and the analogous result follows for $q - r < p(p+1)/2$. Consider, for example, the factor analysis model (1.1). Here $q = pk + p$ and $r = k(k-1)/2$ equality constraints may be imposed (e.g., Lawley and Maxwell (1971)). Therefore if

$$(1.5) \quad pk + p - k(k-1)/2 < p(p+1)/2$$

or equivalently

$$(1.6) \quad (p-k)(p-k+1)/2 > p$$

then the factor analysis model cannot be solved exactly for almost every S (see Shapiro (1982), Shapiro [in press](a)). If an exact solution is impossible one should think about the approximation of S by a matrix of the form $\hat{\Sigma} = \Sigma(\hat{\theta})$. Then some measure of similarity between S and $\hat{\Sigma}$ has to be introduced. Usually it is done by defining a scalar-valued function $F(S, \hat{\Sigma})$ which gives an indication of the discrepancy between the sample covariance matrix S and the fitted matrix $\hat{\Sigma}$. Following Browne (1982) the function $F(S, \hat{\Sigma})$ will be called a discrepancy function. Then for a given S the estimate $\hat{\theta}$ is chosen in such a manner as to minimize $F(S, \Sigma(\hat{\theta}))$, i.e.,

$$(1.7) \quad F(S, \Sigma(\hat{\theta})) = \min_{\theta \in \Omega} F(S, \Sigma(\theta))$$

The long standing tradition of maximum likelihood leads to the following discrepancy function (see Lawley and Maxwell (1971))

$$(1.8) \quad F(S, \hat{\Sigma}) = \log|\hat{\Sigma}| - \log|S| + \text{tr}(S\hat{\Sigma}^{-1}) - p.$$

An attraction of this function is that the standard theory of maximum likelihood may be applied (e.g., Kendall and Stuart (1979), sections 18 and 24).

An alternative approach is based on the generalized least squares (GLS) discrepancy function

$$(1.9) \quad F(S, \hat{\Sigma}) = 2^{-1} \text{tr}[(S - \hat{\Sigma})V]^2$$

where V is a positive definite matrix, possibly depending on S and $\hat{\Sigma}$. For the factor analysis model (1.1) and $V = I$ it turns out to be the usual least squares method (Harman and Jones (1966)). A comprehensive study of the GLS method has been given by Browne (1974). He showed that if V converges in probability to Σ^{-1} (for example $V = S^{-1}$) then the GLS estimators are asymptotically equivalent to the maximum likelihood estimators (see also Jöreskog and Goldberger (1972), Anderson (1973), Lee and Bentler (1980)).

The main object of this paper is to present a unified approach to the theory of covariance structures. Some well-known results as well as new ones will be obtained in the general framework. Because the

maximum likelihood function (1.8) may be considered as a particular case of the discrepancy function, certain similarities between the presented results as well as methods and classical ones of the ML theory will not escape the reader (cf., Wald (1943), Aitchison and Silvey (1958)). However, the main difference is that in covariance structure analysis the sample information is already summarized in the sample covariance matrix. Here the $p(p+1)/2$ distinct elements of $n^2(S-\Sigma)$ are known to have an asymptotic multivariate normal distribution with zero mean. Therefore we may consider the discrepancy function as some sort of mechanism which transfers the asymptotic properties of S through the minimization procedure to the parameter estimator $\hat{\theta}$. The following features of covariance structures theory will be essential for us:

(1) The asymptotic distribution of $n^2(\underline{s} - \underline{\sigma})$ is multivariate normal with zero means and a covariance matrix Γ (here $\underline{s} = \text{vecs}(S)$ and $\underline{\sigma} = \text{vecs}(\Sigma)$, where $\text{vecs}(S)$ denotes the $p(p+1)/2 \times 1$ vector formed from the elements above and including the diagonal of a symmetric matrix S taken columnwise).

(2) The discrepancy function $F(S, \Sigma(\theta))$ is a function of two vector variables $\underline{s}$ and $\underline{\theta}$.

Therefore we consider a real-valued function $f(\underline{x}, \underline{\theta})$ of two vector variables $\underline{x} \in R^m$ and $\underline{\theta} \in \Omega \subseteq R^q$. In particular it may be $\underline{x} = \text{vecs}(S)$ and $f(\underline{x}, \underline{\theta}) = F(S, \Sigma(\theta))$. Then the so-called min-function

$$(1.10) \quad f_*(\underline{x}) = \inf_{\underline{\theta} \in \Omega} f(\underline{x}, \underline{\theta})$$

or the max-function

$$(1.11) \quad f^*(\underline{x}) = \sup_{\underline{\theta} \in \Omega} f(\underline{x}, \underline{\theta})$$

will be of interest. Also we are interested in a value $\hat{\underline{\theta}} = \underline{\theta}(\underline{x})$ of the parameter vector $\underline{\theta}$ where the minimum (or maximum) is attained. In sections 2, 3 and 4 the analytical properties (continuity, differentiability) of the functions $f_*(\underline{x})$ and $\underline{\theta}(\underline{x})$ will be studied. In these sections $\underline{x}$ will be treated as a mathematical vector variable. The main development

will be given in section 4 where, under the identification assumption, the first and second order derivatives will be obtained. In section 3 we summarize some results and ideas from the recently developed theory of non-smooth optimization. We need these results mainly for the case of non-identification and if not interested, the reader may omit this section. In section 5 and afterwards $\underline{x}$ will be considered as a random vector such that $n^{1/2}(\underline{x} - \underline{x}_0)$, where usually $\underline{x}_0 = E(\underline{x})$, is asymptotically multinormal with zero means and some covariance matrix Γ . Then the mini-max calculus of sections 2, 3 and 4 will be applied to obtain the asymptotic distribution properties of $f_*(\underline{x})$ and $\hat{\theta} = \theta(\underline{x})$. The general results of section 5 will be demonstrated in sections 6 and 7 (the GLS method). For example, the asymptotic distribution of the test statistic under an alternative hypothesis will be obtained. Finally in section 8 we are willing to demonstrate that a wider range of problems than the covariance structure analysis may be treated by the developed theory. The considered example of so-called minimum trace factor analysis, may be not too important on its own, but nicely demonstrates the potential possibilities of the theory.

2. CONTINUOUS PROPERTIES OF MIN-FUNCTIONS

Throughout this section we assume that $f(\underline{x}, \theta)$ is a continuous function of $\underline{x}$ for every $\theta \in \Omega$. In spite of the continuity assumption the min-function $f_*(\underline{x})$ need not be continuous. Consider for example

$$(2.1) \quad f(x, \theta) = \exp(-\theta^2 x^2)$$

with $\Omega = \mathbb{R}$.

It can be easily verified that the min-function $f_*(x)$ is

$$(2.2) \quad f_*(x) = \begin{cases} 0, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

which is not continuous at zero.

To ensure the continuity of $f_*(\underline{x})$ further assumptions will be necessary. In this regard the following theorem is most general (see, e.g.,

Pshenichnyi (1971), p.71).

Theorem 2.1

Let Ω be a compact topological space⁽¹⁾ and $f(\underline{x}, \underline{\theta})$ be continuous with respect to $\underline{x}$ and $\underline{\theta}$. Then $f_*(\underline{x})$ is continuous.

Let us define by $\Omega(\underline{x})$ the set of points in Ω at which $f(\underline{x}, \cdot)$ attains its minimum, i.e.

$$(2.3) \quad \Omega(\underline{x}) = \{ \underline{\theta} \in \Omega : f_*(\underline{x}) = f(\underline{x}, \underline{\theta}) \}.$$

If the conditions of Theorem 2.1 hold then $\Omega(\underline{x})$ is a non-empty compact subset of Ω for all $\underline{x}$ and the next theorem follows (e.g. Pshenichnyi (1971), p.72).

Theorem 2.2

Let the conditions of Theorem 2.1 hold. Then for any $\underline{x}_0 \in \mathbb{R}^m$ and every neighbourhood $\omega \subseteq \Omega$ of the set $\Omega(\underline{x}_0)$, there exists a neighbourhood $N(\underline{x}_0)$ of $\underline{x}_0$ such that

$$(2.4) \quad \Omega(\underline{x}) \subseteq \omega$$

whenever $\underline{x} \in N(\underline{x}_0)$.

Definition 2.1

A parameter $\underline{\theta}$ will be said to be conditionally identified (c-identified) at a point $\underline{\theta}_0$ in a parameter space Ω given $\underline{x}_0$ and $f(\underline{x}, \underline{\theta})$ if $\Omega(\underline{x}_0) = \{\underline{\theta}_0\}$, i.e. the set $\Omega(\underline{x}_0)$ contains only one point $\underline{\theta}_0$ (such a set is called a singleton). The parameter $\underline{\theta}$ will be said to be locally c-identified at $\underline{\theta}_0$ given $\underline{x}_0$ if $f(\underline{x}_0, \underline{\theta}) > f(\underline{x}_0, \underline{\theta}_0)$ for every $\underline{\theta}, \underline{\theta} \neq \underline{\theta}_0$, in some neighbourhood of $\underline{\theta}_0$.

(1) If Ω is a subset of $\mathbb{R}^q$ then this simply means that Ω is closed and bounded.

We remark that for $\underline{x} = \text{vecs}(S)$ and $f(\underline{x}, \underline{\theta}) := F(S, \Sigma(\underline{\theta}))$ the conditional identification depends on a choice of the discrepancy function F and generally speaking differs from the usual one: $\Sigma(\underline{\theta}_0) = \Sigma(\underline{\theta}_1)$, $\underline{\theta}_0, \underline{\theta}_1 \in \Omega$ imply that $\underline{\theta}_0 = \underline{\theta}_1$. However, under mild assumptions on F (see e.g., Browne (1982)) they coincide given $S = \Sigma(\underline{\theta}_0)$.

Theorem 2.3

Let $\underline{\theta}$ be c-identified at $\underline{\theta}_0$ given $\underline{x}_0$ and the conditions of Theorem 2.1 hold. Then for any sequence $\underline{x}_i \rightarrow \underline{x}_0$ and $\underline{\theta}_i \in \Omega(\underline{x}_i)$, $i=1,2,\dots$, we have $\underline{\theta}_i \rightarrow \underline{\theta}_0$. In particular, if $\underline{\theta}$ is c-identified given $\underline{x}$ in some neighbourhood of $\underline{x}_0$ we can consider the function $\underline{\theta}(\underline{x})$ defined by $\Omega(\underline{x}) = \{\underline{\theta}(\underline{x})\}$. It follows that $\underline{\theta}(\underline{x})$ is continuous at $\underline{x}_0$.

Proof Follows from Theorem 2.2. ■

In many respects the compactness assumption about the parameter space Ω is inconvenient. It may be replaced by the following condition but cannot be thrown out completely.

Definition 2.2

We say that the level condition holds at a point $\underline{x}_0$ if there exist:

a compact subspace Ω^* of Ω , a number $a > f_*(\underline{x}_0)$ and a neighbourhood $N(\underline{x}_0)$ of $\underline{x}_0$ such that

$$(2.4) \quad \{\underline{\theta} \in \Omega: f(\underline{x}, \underline{\theta}) \leq a\} \subseteq \Omega^*$$

whenever $\underline{x} \in N(\underline{x}_0)$. ■

It is clear that for $\underline{x}$ in $N(\underline{x}_0)$ the minimization is actually done in the compact space Ω^* and therefore the previous results on continuity at $\underline{x}_0$ will be valid.

Another approach to the continuity problem is to impose some conditions on the functions $f_{\underline{\theta}}(\underline{x}) = f(\underline{x}, \underline{\theta})$, $\underline{\theta} \in \Omega$. The two following conditions are most important.

Definition 2.3

A function $g(\underline{x})$ defined on a convex set $D \subseteq \mathbb{R}^m$ is said to be convex

$$(2.5) \quad g(\alpha \underline{x} + (1 - \alpha)\underline{y}) \leq \alpha g(\underline{x}) + (1 - \alpha)g(\underline{y})$$

for every $\underline{x}, \underline{y} \in D$ and every $\alpha \in [0, 1]$. A function $g(\underline{x})$, $\underline{x} \in \mathbb{R}^m$, is said to be locally Lipschitz if for each bounded subset B of $\mathbb{R}^m$ there exists a positive constant K such that

$$(2.6) \quad |g(\underline{x}) - g(\underline{y})| \leq K \|\underline{x} - \underline{y}\|$$

whenever $\underline{x}, \underline{y} \in B$.

It is clear that a locally Lipschitz function is continuous and it is known (e.g., Rockafellar (1970)) that a convex function defined on $\mathbb{R}^m$ is locally Lipschitz and therefore continuous.

Now let the function $f_\theta(\underline{x})$ be convex for every $\theta \in \Omega$ and the max-function $f^*(\underline{x})$ be finite for all $\underline{x}$. Then $f^*(\underline{x})$ is convex, i.e. the maximum of convex functions is convex (e.g., Rockafellar (1970)). Similarly, if the functions $f_\theta(\underline{x})$ are uniformly locally Lipschitz (i.e. the constant K may be chosen independently of θ) then the max-function $f^*(\underline{x})$ and the min-function $f_*(\underline{x})$ are locally Lipschitz.

3. FIRST ORDER DIFFERENTIAL PROPERTIES OF MIN(MAX)-FUNCTIONS

Unfortunately the classical approach of linear approximations (differentiability) does not work very well in minimax problems. Consider for example the absolute value function $|x| = \max\{-x, x\}$. We see that the max-function $|x|$ is not differentiable at zero and this is due to the fact that $x' = 1$ while $(-x)' = -1$ at $x=0$. Therefore we should replace the smoothness assumptions by more general concepts. In this connection the directional derivative approach will be most useful.

Definition 3.1

The directional derivative of a function g at $\underline{x}$ in a direction $\underline{y}$ is defined to be the limit

$$(3.1) \quad g'(\underline{x}; \underline{y}) = \lim_{t \rightarrow 0} \frac{g(\underline{x} + t\underline{y}) - g(\underline{x})}{t}$$

if it exists.

Of course, if g is actually differentiable at $\underline{x}$ then the directional derivative $g'(\underline{x}; \underline{y})$ exists for every $\underline{y}$ and one has

$$(3.2) \quad g'(\underline{x}; \underline{y}) = \underline{y}' \nabla g(\underline{x})$$

where $\nabla g(\underline{x}) = \left(\frac{\partial g(\underline{x})}{\partial x_1}, \dots, \frac{\partial g(\underline{x})}{\partial x_m} \right)'$ denotes the gradient vector of

g at $\underline{x}$ (also will be denoted by $\frac{\partial g(\underline{x})}{\partial \underline{x}}$).

We note that the directional derivative $g'(\underline{x}; \underline{y})$ is positively homogeneous with respect to $\underline{y}$, i.e. $g'(\underline{x}; t\underline{y}) = t g'(\underline{x}; \underline{y})$ for every $t \geq 0$. Especially important to us will be functions which have a convex (or concave) with respect to $\underline{y}$ directional derivative $g'(\underline{x}; \underline{y})$. Before giving an exact definition of these, so-called quasi-differentiable, functions we need the following definition.

Definition 3.2

The function $s(\cdot; A)$ defined by

$$(3.3) \quad s(\underline{y}; A) = \sup_{\underline{z} \in A} \underline{y}' \underline{z}$$

where $\underline{y} \in \mathbb{R}^m$ and A is a subset of $\mathbb{R}^m$, is called the support function of A .

In the following proposition we summarize basic properties of support functions (see, e.g., Rockafellar (1970)).

Proposition 3.1

(a) The support function $s(\underline{y}; A)$ of a bounded set A is a convex, continuous and positively homogeneous function.

(b) Conversely, if a function $h(\underline{y})$ is convex and positively homogeneous then it is the support function of a certain convex, compact set A . This set is unique and given by

$$A = \{ \underline{z} \in \mathbb{R}^m : \underline{z}'\underline{y} \leq h(\underline{y}) \text{ for all } \underline{y} \in \mathbb{R}^m \}.$$

$$(c) \quad s(\cdot; A) \equiv s(\cdot; \overline{\text{co}\{A\}})$$

where $\text{co}\{A\}$ denotes the convex hull of A and $\overline{\text{co}\{A\}}$ is its topological closure in $\mathbb{R}^m$.

(d) If A_1, A_2 are convex, compact sets then

$s(\cdot; A_1) \equiv s(\cdot; A_2)$ if and only if $A_1 = A_2$, and the inequality $s(\cdot; A_1) \geq s(\cdot; A_2)$ holds if and only if $A_1 \supseteq A_2$.

(e) For $\lambda, \mu \geq 0$ it follows that

$$s(\cdot; \lambda A_1 + \mu A_2) \equiv \lambda s(\cdot; A_1) + \mu s(\cdot; A_2).$$

Now, following Pshenichnyi (1971), we introduce the concept of quasi-differentiability, which has been found to be very fruitful in minimax problems.

Definition 3.3

A function $g: \mathbb{R}^m \rightarrow \mathbb{R}$ is said to be upper quasi-differentiable (u.q.d.) at $\underline{x}_0$ if the directional derivative $g'(\underline{x}_0; \underline{y})$ exists for every $\underline{y} \in \mathbb{R}^m$ and is convex in $\underline{y}$, i.e., there is a convex, compact set $A \subseteq \mathbb{R}^m$ such that

$$(3.4) \quad g'(\underline{x}_0; \cdot) \equiv s(\cdot; A)$$

The set A will be called the subgradient of g at $\underline{x}_0$ and denoted by $\partial g(\underline{x}_0)$.

A function g is said to be lower quasi-differentiable (l.q.d.) if $-g$ is u.q.d.

By the definition of the directional derivative, equality (3.4) can be written in the following form

$$(3.5) \quad g(\underline{x}_0 + \underline{y}) - g(\underline{x}_0) = s(\underline{y}; A) + \epsilon(\underline{y})$$

where $\epsilon(\underline{y})/t \rightarrow 0$ as $t \rightarrow 0$ for any given direction $\underline{y}$. We may strengthen Definition 3.3 by saying that g is *strongly* u.q.d. at $\underline{x}_0$ if $\epsilon(\underline{y})/|\underline{y}| \rightarrow 0$ as $|\underline{y}| \rightarrow 0$. However, it can be shown that for locally Lipschitz functions the definitions of strongly u.q.d. and u.q.d. coincide.

The equality (3.5) may be viewed as the first-order approximation of g at $\underline{x}_0$ by the support function $s(\cdot; \partial g(\underline{x}_0))$. It can be seen from the definition that $s(\underline{y}; \partial g(\underline{x}_0))$ is linear in $\underline{y}$ if (and only if) the set $\partial g(\underline{x}_0)$ contains only one vector. Therefore the u.q.d. function g is differentiable at $\underline{x}_0$ in the usual sense if (and only if) the subgradient $\partial g(\underline{x}_0)$ contains only one vector, which clearly will be the gradient of g at $\underline{x}_0$.

The usefulness of the quasi-differentiability concept lies in the fact that many max-functions (min-functions) and in particular convex functions are u.q.d. (l.q.d.). In the following theorem we summarize some results of this sort.

Theorem 3.1

Let $f_{\theta}(\underline{x}) = f(\underline{x}, \theta)$ be a function of two variables $\underline{x} \in \mathbb{R}^m$ and $\theta \in \Omega$ where Ω is a compact topological space. Then $f^*(\underline{x})$ is u.q.d. at $\underline{x}_0$ and

$$(3.6) \quad \partial f^*(\underline{x}_0) = \overline{\text{co}}\{ \bigcup_{\theta \in \Omega(\underline{x}_0)} \partial f(\underline{x}_0, \theta) \}$$

if one of the following conditions holds:

(a) $f(\underline{x}, \theta)$ is differentiable in $\underline{x}$ for every $\theta \in \Omega$ and the gradient mapping $\nabla f(\underline{x}, \theta)$ is continuous in $\underline{x}$ and θ .

(Danskin (1967)).

(b) $f(\underline{x}, \theta)$ is convex in $\underline{x}$ for every $\theta \in \Omega$ and continuous in θ for every $\underline{x}$ (e.g., Ioffe and Tihomirov (1979), p.201).

(c) $f(\underline{x}, \theta)$ is continuous in $\underline{x}$ and θ , u.q.d. at $\underline{x}_0$ for every $\theta \in \Omega$ and for each $\underline{y} \in \mathbb{R}^m$:

$$f_{\theta}(\underline{x}_0 + t\underline{y}) - f_{\theta}(\underline{x}_0) = f'_{\theta}(\underline{x}_0; t\underline{y}) + t \epsilon(t, \theta)$$

where $\epsilon(t, \theta) \rightarrow 0$ uniformly in θ as $t \rightarrow 0$ (Pshenichnyi (1971)).

Note that the compactness of Ω can be replaced by the level condition at $\underline{x}_0$ (see Definition 2.2).

Corollary 3.1

Let Ω be a compact topological space and condition (a) of Theorem 3.1 holds. Then $f^*(\underline{x})$ is differentiable at $\underline{x}_0$ if and only if the gradients $\nabla f_{\theta}(\underline{x}_0)$ are identical for all $\theta \in \Omega(\underline{x}_0)$. In particular,

$f^*(\underline{x}_0)$ is differentiable at $\underline{x}_0$ and $\nabla f^*(\underline{x}_0) = \nabla f_{\underline{\theta}_0}(\underline{x}_0)$ if the parameter $\underline{\theta}$ is c-identified at $\underline{\theta}_0$ given $\underline{x}_0$.

We end this section by the following example. Let S be a $p \times p$ symmetric matrix and $\lambda_1(S) \geq \lambda_2(S) \geq \dots \geq \lambda_p(S)$ denote its eigenvalues. Define $L(S)$ by

$$(3.7) \quad L(S) = \lambda_1(S) + \dots + \lambda_k(S)$$

where $1 \leq k \leq p$. It is known that $L(S)$ can be represented as follows (e.g., Bellman (1960)).

$$(3.8) \quad L(S) = \max_{E \in \Omega} f(S, E)$$

where $f(S, E) = \text{tr } E'SE$ and Ω is the set of such $p \times k$ matrices E that $E'E = I_k$. Furthermore the maximum in (3.8) is attained at E if and only if the column vectors of E are eigenvectors of S corresponding to eigenvalues $\lambda_1(S), \dots, \lambda_k(S)$. The function $f(S, E)$, which may be written as

$$(3.9) \quad f(S, E) = \text{tr } SEE'$$

is linear and therefore convex and differentiable in S . Its gradient (in S) is given by

$$(3.10) \quad \nabla f(S, E) = EE'$$

Now Theorem 3.1 (condition (b)) can be applied, where the max-representation of $L(S)$ is given by (3.8). We obtain that $L(S)$ is convex (as the maximum of convex functions) and therefore u.q.d.. And

$$(3.11) \quad \partial L(S) = \text{co}\{EE' : E'E = I_k ; SE = EA\}$$

where A is the $k \times k$ diagonal matrix with $\lambda_1(S), \dots, \lambda_k(S)$ diagonal elements.

In particular it follows that L is differentiable at S if and only if $\lambda_{k+1}(S) \neq \lambda_k(S)$.

4. SECOND-ORDER DERIVATIVES OF MIN(MAX)-FUNCTIONS

Throughout this section we assume that the parameter space Ω is a compact subset of R^q (or the level condition holds at a considered point $\underline{x}_0$) and $f(\underline{x}, \underline{\theta}) : R^m \times \Omega \rightarrow R$ is a twice continuously differentiable function of $m + q$ variables $x_1, \dots, x_m, \theta_1, \dots, \theta_q$. The Hessian matrices of second order partial derivatives

$\frac{\partial^2 f}{\partial \theta \partial \theta}$, $\frac{\partial^2 f}{\partial \theta \partial x}$, $\frac{\partial^2 f}{\partial x \partial x}$ at a point $(\underline{x}, \underline{\theta})$ will be denoted by

$H_{\theta\theta}(\underline{x}, \underline{\theta})$, $H_{\theta x}(\underline{x}, \underline{\theta})$, $H_{xx}(\underline{x}, \underline{\theta})$ respectively.

It is well known that if for a given $\underline{x} = \underline{x}_0$ the function f attains its minimum (maximum) in $\underline{\theta}$ at an interior point $\underline{\theta}_0$ of Ω , then the following (generally nonlinear) equations

$$(4.1) \quad \frac{\partial f}{\partial \theta_i} = 0, \quad i = 1, \dots, q$$

hold at $(\underline{x}_0, \underline{\theta}_0)$, (necessary conditions). Furthermore the Hessian matrix $H_{\theta\theta}(\underline{x}_0, \underline{\theta}_0)$ must be nonnegative definite. On the other hand if the first order partial derivatives (4.1) are zero at $(\underline{x}_0, \underline{\theta}_0)$ and $H_{\theta\theta}(\underline{x}_0, \underline{\theta}_0)$ is positive definite then $\underline{\theta}_0$ is a local minimum solution (sufficient condition). In our terminology it means that the parameter $\underline{\theta}$ is locally α -identified at $\underline{\theta}_0$ given $\underline{x}_0$.

The $q \times (m + q)$ matrix $[H_{\theta x}, H_{\theta\theta}]$ can be considered as the Jacobian matrix of the gradient mapping $\frac{\partial f}{\partial \underline{\theta}} = (\frac{\partial f}{\partial \theta_1}, \dots, \frac{\partial f}{\partial \theta_q})'$. If $H_{\theta\theta}$ is nonsingular (positive definite) at $(\underline{x}_0, \underline{\theta}_0)$ then the implicit function theorem (e.g., Apostol (1974)) can be applied to equations (4.1). It follows that equations (4.1) have a unique solution $\underline{\theta}(\underline{x})$ for every $\underline{x}$ in some neighbourhood of $\underline{x}_0$ and $\underline{\theta}(\underline{x}) \rightarrow \underline{\theta}_0$ as $\underline{x} \rightarrow \underline{x}_0$. The elements of $H_{\theta\theta}$

depend continuously on $(\underline{x}, \underline{\theta})$. Therefore positive definiteness of $H_{\theta\theta}$ at $(\underline{x}_0, \underline{\theta}_0)$ implies positive definiteness in some neighbourhood of $(\underline{x}_0, \underline{\theta}_0)$. It follows that if $H_{\theta\theta}(\underline{x}_0, \underline{\theta}_0)$ is positive definite then the solution $\underline{\theta}(\underline{x})$ of (4.1) is the unique minimum in some neighbourhood of $\underline{\theta}_0$ for every $\underline{x}$ sufficiently close to $\underline{x}_0$. Together with Corollary 2.1 this implies the following result.

Theorem 4.1

Suppose that for a given $\underline{x} = \underline{x}_0$ the function f attains its minimum at an interior point $\underline{\theta}_0$ of Ω and that $H_{\theta\theta}(\underline{x}_0, \underline{\theta}_0)$ is nonsingular. Then there exists a neighbourhood Ω_0 of $\underline{\theta}_0$, $\Omega_0 \subseteq \Omega$, such that $\underline{\theta}$ is c-identified in Ω_0 at $\underline{\theta}^* = \underline{\theta}(\underline{x})$ given $\underline{x}$ in some neighbourhood of $\underline{x}_0$ and $\underline{\theta}(\underline{x}) \rightarrow \underline{\theta}_0 = \underline{\theta}(\underline{x}_0)$ as $\underline{x} \rightarrow \underline{x}_0$. Furthermore, if in addition $\underline{\theta}$ is c-identified in Ω at $\underline{\theta}_0$ given $\underline{x}_0$ then $\underline{\theta}$ is c-identified in Ω at $\underline{\theta}(\underline{x})$ for every $\underline{x}$ sufficiently close to $\underline{x}_0$.

So far we used the implicit function theorem to establish the existence and uniqueness of the solution $\underline{\theta}(\underline{x})$ of equations (4.1) and its continuity in $\underline{x}$. Now we apply the implicit function theorem to obtain an explicit expression for derivatives of $\underline{\theta}(\underline{x})$ and then of $f_*(\underline{x})$.

Theorem 4.2

Let the following regularity conditions hold:

- (a) The parameter $\underline{\theta}$ is c-identified in Ω at $\underline{\theta}_0$ given $\underline{x}_0$.
- (b) $\underline{\theta}_0$ is an interior point of Ω .
- (c) The $q \times q$ Hessian matrix $H_{\theta\theta}(\underline{x}_0, \underline{\theta}_0)$ is nonsingular.

Then

(i) The vector-valued function $\underline{\theta}(\underline{x})$ defined in Theorem 4.1 is continuously differentiable at $\underline{x}_0$ and the $q \times m$ Jacobian matrix $\frac{\partial \underline{\theta}}{\partial \underline{x}}$ of $\underline{\theta}(\underline{x})$ at $\underline{x}_0$ is

$$(4.2) \quad \frac{\partial \underline{\theta}}{\partial \underline{x}} = -H_{\theta\theta}^{-1} H_{\theta x}$$

(ii) The function $f_*(\underline{x})$ is twice continuously differentiable at $\underline{x}_0$ and

$$(4.3) \quad \nabla f_*(\underline{x}_0) = \nabla f_{\theta_0}(\underline{x}_0)$$

$$(4.4) \quad \frac{\partial^2 f_*}{\partial \underline{x} \partial \underline{x}} = H_{xx} - H_{\theta x} H_{\theta\theta}^{-1} H_{\theta x}$$

where $\frac{\partial^2 f_*}{\partial \underline{x} \partial \underline{x}}$ is the Hessian of f_* at $\underline{x}_0$.

(All Hessian matrices are taken at $(\underline{x}_0, \underline{\theta}_0)$.)

Proof

We know that $\underline{\theta}(\underline{x})$ is the solution of (4.1). Therefore direct application of the implicit function theorem to equations (4.1) yields (4.2). Now

$$(4.5) \quad f_*(\underline{x}) = f(\underline{x}, \underline{\theta}(\underline{x})).$$

and by the chain rule we have

$$(4.6) \quad \left. \frac{\partial f(\underline{x}, \underline{\theta}(\underline{x}))}{\partial \underline{x}} \right|_{\underline{x}=\underline{x}_0} = \left(\frac{\partial f}{\partial \underline{x}} + \frac{\partial f}{\partial \underline{\theta}} \frac{\partial \underline{\theta}}{\partial \underline{x}} \right) \bigg|_{\substack{\underline{x}=\underline{x}_0 \\ \underline{\theta}=\underline{\theta}_0}}$$

But $\frac{\partial f(\underline{x}_0, \underline{\theta}_0)}{\partial \underline{\theta}} = 0$ and hence (4.6) implies (4.3). Note that the first order derivatives equality (4.3) also follows from Theorem 3.1 (Corollary 3.1). The Hessian matrix of $f_*(\underline{x})$ can be easily calculated

$$(4.7) \quad \frac{\partial^2 f}{\partial \underline{x} \partial \underline{x}'} = \frac{\partial^2 f}{\partial \underline{x} \partial \underline{x}'} + \left(\frac{\partial \theta}{\partial \underline{x}'} \right)' \left(\frac{\partial^2 f}{\partial \theta \partial \underline{x}'} \right)$$

Substituting expression (4.2) of $\frac{\partial \theta}{\partial \underline{x}'}$ in (4.7) we obtain (4.4). ■

In Theorem 4.2 we derived the first-order derivatives of $\underline{\theta}(\underline{x})$ and second-order derivatives of $f_*(\underline{x})$ assuming that the function $f(\underline{x}, \underline{\theta})$ is twice continuously differentiable. In general the implicit function theorem implies that $\underline{\theta}(\underline{x})$ is $(n-1)$ -order differentiable and $f_*(\underline{x})$ is n -order differentiable if the function $f(\underline{x}, \underline{\theta})$ admits n -order continuous derivatives (i.e., the gradient mapping $\frac{\partial f}{\partial \underline{\theta}}$ admits $(n-1)$ -order continuous derivatives). Unfortunately the high order derivatives take too complicated a form to be written explicitly. Using the matrix differentiable calculus, summarized by Nel (1980, section 3), we calculate in the following theorem the second-order derivatives of $\underline{\theta}(\underline{x})$.

Theorem 4.3

Let the regularity conditions of Theorem 4.2 hold and $f(\underline{x}, \underline{\theta})$ be three times continuously differentiable. Then $\underline{\theta}(\underline{x})$ is twice continuously differentiable at $\underline{x}_0$ and

$$(4.8) \quad \frac{\partial^2 \underline{\theta}}{\partial \underline{x}_j \partial \underline{x}_i} = -H^{-1} \left[\frac{\partial^3 f}{\partial \underline{x}_j \partial \underline{x}_i \partial \underline{\theta}} + \frac{\partial H}{\partial \underline{x}_i} \frac{\partial \underline{\theta}}{\partial \underline{x}_j} + \frac{\partial H}{\partial \underline{x}_j} \frac{\partial \underline{\theta}}{\partial \underline{x}_i} + \frac{\partial H}{\partial \underline{\theta}'} \left(\frac{\partial \underline{\theta}}{\partial \underline{x}_j} \otimes \frac{\partial \underline{\theta}}{\partial \underline{x}_i} \right) \right]$$

where $H = H_{\theta\theta'}(\underline{x}_0, \underline{\theta}_0)$, $\frac{\partial \underline{\theta}}{\partial \underline{x}_i} = -H^{-1} \frac{\partial^2 f}{\partial \underline{x}_i \partial \underline{\theta}}$ by (4.2) and

$\frac{\partial H}{\partial \underline{\theta}'} = \left[\frac{\partial H_{ts}}{\partial \underline{\theta}'} \right]$ is a $q \times q^2$ matrix (see Nel (1980), Definition 5.2.1).

Proof

Differentiation of (4.1) by $\underline{x}_i$ gives us

$$(4.9) \quad \frac{\partial^2 f}{\partial \underline{x}_i \partial \underline{\theta}} + \frac{\partial^2 f}{\partial \underline{\theta} \partial \underline{\theta}'} \frac{\partial \underline{\theta}}{\partial \underline{x}_i} = 0$$

and then differentiation of (4.9) by $\underline{x}_j$ yields

$$(4.10) \quad \frac{\partial^3 f}{\partial x_j \partial x_i \partial \theta} + \frac{\partial^3 f}{\partial \theta \partial x_i \partial \theta} \frac{\partial \theta}{\partial x_j} + \left[\frac{\partial}{\partial x_j} \left(\frac{\partial^2 f}{\partial \theta \partial \theta} \right) \right] \frac{\partial \theta}{\partial x_i} + \frac{\partial^2 f}{\partial \theta \partial \theta} \frac{\partial^2 \theta}{\partial x_j \partial x_i} = 0$$

Now

$$(4.11) \quad \frac{\partial H_{ts}(x, \theta(x))}{\partial x_j} = \frac{\partial H_{ts}}{\partial x_j} + \frac{\partial H_{ts}}{\partial \theta^t} \frac{\partial \theta}{\partial x_j}$$

where H_{ts} is the (t,s) -element of H . Therefore

$$(4.12) \quad \left[\frac{\partial H}{\partial x_j} \frac{\partial \theta}{\partial x_i} \right]_t = \sum_s \left(\frac{\partial H_{ts}}{\partial x_j} + \frac{\partial H_{ts}}{\partial \theta^t} \frac{\partial \theta}{\partial x_j} \right) \frac{\partial \theta_s}{\partial x_i}$$

Expression (4.12) can be written in the following matrix form

$$(4.13) \quad \left[\frac{\partial}{\partial x_j} \left(\frac{\partial^2 f}{\partial \theta \partial \theta} \right) \right] \frac{\partial \theta}{\partial x_i} = \frac{\partial H}{\partial x_j} \frac{\partial \theta}{\partial x_i} + \frac{\partial H}{\partial \theta^t} \left(\frac{\partial \theta}{\partial x_j} \otimes \frac{\partial \theta}{\partial x_i} \right)$$

Substituting (4.13) into (4.10) we obtain (4.8).

Often the parameter vector θ is subject to equality constraints given by r equations $g_i(x, \theta) = 0$, $i=1, \dots, r$. Although in applications the functions g_i usually depend only on θ we consider here the more general case, when g_i can also depend on x . The min-function $f_*(x)$ will be defined as follows

$$(4.14) \quad f_*(x) = \min_{\theta \in \Omega} f(x, \theta) \quad \text{subject to} \quad g(x, \theta) = 0$$

where $g(x, \theta) = (g_1(x, \theta), \dots, g_r(x, \theta))'$ will be a vector-valued function representing the equality constraints. Throughout we assume that the functions

$g_i(\underline{x}, \underline{\theta})$, $i=1, \dots, r$ are twice continuously differentiable in $\underline{x}$ and $\underline{\theta}$. The Lagrange function of $m + q + r$ variables will be defined as follows

$$(4.15) \quad L(\underline{x}, \underline{\theta}, \underline{\lambda}) = f(\underline{x}, \underline{\theta}) + \underline{g}'(\underline{x}, \underline{\theta}) \underline{\lambda},$$

where $\underline{\lambda} = (\lambda_1, \dots, \lambda_r)'$ is a vector of Lagrange multipliers. Then the Lagrange principle consists of the following:

Let $\underline{\theta}_0$ be a local minimizer of the problem (4.14) for a fixed $\underline{x} = \underline{x}_0$. Then there exists a vector $\underline{\lambda}_0$ of Lagrange multipliers such that

$$(4.16) \quad \begin{cases} \frac{\partial f(\underline{x}_0, \underline{\theta}_0)}{\partial \underline{\theta}} + \frac{\partial \underline{g}(\underline{x}_0, \underline{\theta}_0)'}{\partial \underline{\theta}} \underline{\lambda}_0 = 0 \\ \underline{g}(\underline{x}_0, \underline{\theta}_0) = 0 \end{cases}$$

We remark that in order to apply the Lagrange principle a *constraint qualification* must be satisfied. The constraint qualifications, widely known in nonlinear programming, are

(I) The $q \times r$ Jacobian matrix $G_{\theta} = \frac{\partial \underline{g}'}{\partial \underline{\theta}}$ is of full column rank r at $(\underline{x}_0, \underline{\theta}_0)$,

or (Beltrami (1969))

(II) The Jacobian matrix $G_{\theta} = \frac{\partial \underline{g}'}{\partial \underline{\theta}}$ is of constant rank for each $\underline{\theta}$ in some neighbourhood of $\underline{\theta}_0$ and $\underline{x} = \underline{x}_0$.

Let us denote by $\underline{\mu}$ the $(q+r)$ -dimensional column vector $\begin{pmatrix} \underline{\theta} \\ \underline{\lambda} \end{pmatrix}$. Then the $q+r$ equations (4.16) can be expressed in the following form

$$(4.17) \quad \frac{\partial L(\underline{x}, \underline{\mu})}{\partial \underline{\mu}} = 0$$

The equations (4.17) are parallel to equations (4.1). Therefore, the following theorem will be an analogue of Theorem 4.2.

Theorem 4.4

Let the following regularity conditions hold:

(a) The parameter $\underline{\mu}$ is c-identified at $\underline{\mu}_0$ given $\underline{x}_0$, i.e., for $\underline{x} = \underline{x}_0$ there is a unique minimizer $\underline{\theta}_0$ in (4.14) and the Lagrange principle holds with a unique vector $\underline{\lambda}_0$ of Lagrange multipliers.

(b) $\underline{\theta}_0$ is an interior point of Ω .

(c) The $(q+r) \times (q+r)$ Hessian matrix $H_{\mu\mu}(\underline{x}_0, \underline{\mu}_0)$ of $L(\underline{x}, \underline{\mu})$ is nonsingular.

Then

(i) $\underline{\mu}$ is c-identified at $\underline{\mu}^* = \underline{\mu}(\underline{x})$ given $\underline{x}$ in some neighbourhood of $\underline{x}_0$

(ii) The vector-valued function $\underline{\mu}(\underline{x})$ is continuously differentiable at $\underline{x}_0$ and

$$(4.18) \quad \frac{\partial \underline{\mu}(\underline{x}_0)}{\partial \underline{x}'} = -H_{\mu\mu}^{-1} H_{\mu x}'$$

(iii) The function $f_*(\underline{x})$ is twice continuously differentiable at $\underline{x}_0$ and

$$(4.19) \quad \nabla f_*(\underline{x}_0) = \frac{\partial L(\underline{x}_0, \underline{\mu}_0)}{\partial \underline{x}} = \frac{\partial f(\underline{x}_0, \underline{\theta}_0)}{\partial \underline{x}} + \frac{\partial g(\underline{x}_0, \underline{\theta}_0)'}{\partial \underline{x}} \underline{\lambda}_0$$

$$(4.20) \quad \frac{\partial^2 f_*(\underline{x}_0)}{\partial \underline{x} \partial \underline{x}'} = H_{xx}' - H_{\mu x}' H_{\mu\mu}^{-1} H_{\mu x}'$$

(Here all Hessians are of the Lagrange function $L(\underline{x}, \underline{\mu})$ and taken at $(\underline{x}_0, \underline{\mu}_0)$.)

Note that the Hessian matrix $H_{\mu\mu}'$ can be partitioned as follows:

$$(4.21) \quad H_{\mu\mu}' = \left[\begin{array}{c|c} H_{\theta\theta}' & G_{\theta} \\ \hline G_{\theta}' & 0 \end{array} \right]$$

$$\text{where } H_{\theta\theta}' = \frac{\partial^2 L}{\partial \theta \partial \theta'} = \frac{\partial^2 f}{\partial \theta \partial \theta'} + \sum_{i=1}^r \lambda_i \frac{\partial^2 g_i}{\partial \theta \partial \theta'}$$

and $G_{\theta} = \frac{\partial g}{\partial \theta}$. The nonsingularity of $H_{\mu\mu}'$ implies that G_{θ} has full column rank r (i.e., condition (I) holds). Then from (4.16)

we have $\lambda_0 = -(G_{\theta}' G_{\theta})^{-1} G_{\theta}' (\frac{\partial f}{\partial \theta})$ and thus λ_0 exists and is unique. Therefore in condition (a) of Theorem 4.4 we actually must verify only the uniqueness of the minimizer θ_0 .

Let

$$(4.22) \quad \left[\begin{array}{c|c} H_{\theta\theta}' & G_{\theta} \\ \hline G_{\theta}' & 0 \end{array} \right]^{-1} = \left[\begin{array}{c|c} C_1 & C_2 \\ \hline C_2' & C_3 \end{array} \right]$$

and

$$(4.23) \quad H_{\theta x'} = \frac{\partial^2 L}{\partial \theta \partial x'} = \frac{\partial^2 f}{\partial \theta \partial x'} + \sum_{i=1}^r \lambda_i \frac{\partial^2 g_i}{\partial \theta \partial x'}$$

$$(4.24) \quad H_{xx'} = \frac{\partial^2 L}{\partial x \partial x'} = \frac{\partial^2 f}{\partial x \partial x'} + \sum_{i=1}^r \lambda_i \frac{\partial^2 g_i}{\partial x \partial x'}$$

$$(4.25) \quad G_{x'} = \frac{\partial g}{\partial x'}$$

Then we can write the formulas of Theorem 4.4 in a more explicit form as follows

$$(4.26) \quad \begin{bmatrix} \frac{\partial \theta}{\partial \underline{x}'} \\ \frac{\partial \lambda}{\partial \underline{x}'} \end{bmatrix} = - \begin{bmatrix} C_1 & C_2 \\ C_2' & C_3 \end{bmatrix} \begin{bmatrix} H_{\theta \underline{x}'} \\ G_{\underline{x}'} \end{bmatrix}$$

$$(4.27) \quad \frac{\partial^2 f^*}{\partial \underline{x} \partial \underline{x}'} = H_{\underline{x} \underline{x}'} = \begin{bmatrix} H_{\theta \underline{x}'} \\ G_{\underline{x}'} \end{bmatrix}' \begin{bmatrix} C_1 & C_2 \\ C_2' & C_3 \end{bmatrix} \begin{bmatrix} H_{\theta \underline{x}'} \\ G_{\underline{x}'} \end{bmatrix}$$

(all derivatives and Hessians are taken at $(\underline{x}_0, \theta_0, \lambda_0)$).

In particular, if $\underline{g}$ is independent of $\underline{x}$ then

$$G_{\underline{x}'} = 0, \quad H_{\theta \underline{x}'} = \frac{\partial^2 f}{\partial \theta \partial \underline{x}'}, \quad H_{\underline{x} \underline{x}'} = \frac{\partial^2 f}{\partial \underline{x} \partial \underline{x}'}, \quad \text{and}$$

$$(4.28) \quad \frac{\partial \theta}{\partial \underline{x}'} = -C_1 H_{\theta \underline{x}'}$$

$$(4.29) \quad \frac{\partial \lambda}{\partial \underline{x}'} = -C_2' H_{\theta \underline{x}'}$$

$$(4.30) \quad \frac{\partial^2 f^*}{\partial \underline{x} \partial \underline{x}'} = H_{\underline{x} \underline{x}'} - H_{\theta \underline{x}'}' C_1 H_{\theta \underline{x}'}$$

We have mentioned earlier that a necessary condition for nonsingularity of $H_{\mu\mu'}$ is that G_θ has full column rank r . Furthermore the following result is easy to establish (e.g., Rao (1973)).

Proposition 4.1

If the matrix $H_{\theta\theta'}$ in (4.21) is nonnegative definite then $H_{\mu\mu'}$ is nonsingular if and only if the matrices $[H_{\theta\theta'}; G_\theta]$ and G_θ have rank q and r respectively.

In some cases explicit expressions for the matrices C_1 , C_2 and C_3 can be given. (For typographical brevity we denote in the following proposition $H = H_{\theta\theta}$, and $G = G_\theta$).

Proposition 4.2

Suppose that in the following all required inverses exist. Then $H_{\mu\mu}$ is nonsingular and

$$(a) \quad C_1 = H^{-1} - H^{-1}G(G'H^{-1}G)^{-1}G'H^{-1}$$

$$C_2' = (G'H^{-1}G)^{-1}G'H^{-1}$$

$$C_3 = -(G'H^{-1}G)^{-1}$$

(e.g., Silvey (1970)).

$$(b) \quad C_1 = H_0^{-1} - H_0^{-1}G(G'H_0^{-1}G)^{-1}G'H_0^{-1}$$

$$C_2' = (G'H_0^{-1}G)^{-1}G'H_0^{-1}$$

$$C_3 = I - (G'H_0^{-1}G)^{-1}$$

where $H_0 = H + GG'$

(e.g., Pringle and Rayner (1971)).

(c) Let W be a $q \times (q - r)$ matrix of rank $q - r$ such that $W'G = 0$, then

$$(4.31) \quad H^{-1} - H^{-1}G(G'H^{-1}G)^{-1}G'H^{-1} = W(W'HW)^{-1}W'$$

(Khatrij (1966)) and therefore

$$C_1 = W(W'HW)^{-1}W'.$$

Apart from the explicit expressions Proposition 4.2 gives us various criteria for verification of nonsingularity of $H_{\mu\mu}$. For instance, from (a) (from (b)) follows that $H_{\mu\mu}$ is nonsingular if

$H_{\theta\theta}$ (if $H_0 = H_{\theta\theta} + G_\theta G_\theta'$) is positive definite and G_θ is of rank

If $\underline{g}$ is independent of $\underline{x}$ then we have from (4.28), (4.30) and Proposition 4.2(c):

$$(4.32) \quad \frac{\partial \theta}{\partial \underline{x}} = -W(W'H_{\theta\theta}W)^{-1}WH_{\theta x},$$

$$(4.33) \quad \frac{\partial^2 f}{\partial \underline{x} \partial \underline{x}'} = H_{xx} - H_{\theta x}' W(W'H_{\theta\theta}W)^{-1}WH_{\theta x},$$

where W is defined as in Proposition 4.2(c), the Hessians H_{xx} and $H_{\theta x}$ are of the function f and $H_{\theta\theta}$ is of the Lagrange function L .

Formulas (4.32) and (4.33) are applicable if $W'H_{\theta\theta}W$ is nonsingular and the constraint qualification (I) holds (compare with the unconstrained formulas (4.2) and (4.4)). If the constraint qualification (II) replaces (I) then, in order to retain (4.32) and (4.33), the matrix W must be of order $q \times (q - \ell)$ and of rank $q - \ell$, where $\ell = \text{rank}(G_\theta)$.

5. APPLICATIONS TO ASYMPTOTIC DISTRIBUTION THEORY, BASIC RESULTS

In this section we consider applications of the min-max calculus developed in sections 2,3 and 4 to specific problems of asymptotic distribution theory. The functions $f(\underline{x}, \underline{\theta})$, $f_*(\underline{x})$ and $\underline{\theta}(\underline{x})$ will be defined as earlier and $\hat{\underline{x}}$ will represent a given estimate (based on a sample of size $N = n + 1$) of a true although unknown quantity $\underline{x}_0$. We are concerned with quantities $f_0 = f_*(\underline{x}_0)$ and $\underline{\theta}_0 = \underline{\theta}(\underline{x}_0)$, which we estimate by $\hat{f}_* = f_*(\hat{\underline{x}})$ and $\hat{\underline{\theta}} = \underline{\theta}(\hat{\underline{x}})$. In other words the estimation method considered here chooses $\hat{\underline{\theta}}$ in such a manner as to minimize (in $\underline{\theta}$) the function $f(\underline{x}, \underline{\theta})$ for $\underline{x} = \hat{\underline{x}}$.

By " $\hat{x}$ is a consistent estimate of x_0 " we mean that the corresponding estimator converges in probability to x_0 as $n \rightarrow \infty$.

Theorem 5.1

Let $\hat{x}$ be a consistent estimate of x_0 and let Ω be a compact topological space and $f(x, \theta)$ be continuous in x and θ . Then $\hat{f}_*$ is a consistent estimate of f_0 . Furthermore, if in addition the parameter θ is c-identified in Ω at θ_0 given x_0 then $\hat{\theta}$ (possibly not unique) is a consistent estimate of θ_0 .

Proof

The theorem follows from the continuity Theorems 2.1 and 2.3 (see e.g. Rao (1973), 2c.4(xiii)).

Note that the compactness of Ω can be replaced by the level condition at x_0 .

From now we assume the estimate $\hat{x}$ to be asymptotically normal, i.e., the distribution of $n^{\frac{1}{2}}(\hat{x} - x_0)$ approaches a multivariate normal distribution with zero means and a covariance matrix Γ as $n \rightarrow \infty$. The asymptotic distribution theory of large samples is well known. The following basic result can be found in Rao (1973), p.388, (Delta Theorem).

Theorem 5.2

Let a vector-valued function $e(x) = (e_1(x), \dots, e_k(x))'$ be differentiable at $x = x_0$. Then $n^{\frac{1}{2}}(e(\hat{x}) - e(x_0))$ is asymptotically normal with zero means and covariance matrix $E'E'$, where $E = \frac{\partial e}{\partial x}$ is the Jacobian matrix of e at x_0 .

Following the same line of proof we extend this result in two directions. First we consider a nondifferentiable case. Here and afterwards " $\stackrel{a}{\approx}$ " is read "is asymptotically equal to" and means that the difference between the left and right sides approaches zero in probability as $n \rightarrow \infty$.

Proposition 5.1

Let a function $h(\underline{x})$ be strongly u.q.d. at $\underline{x}_0$. Then

(5.1) $n^{\frac{1}{2}}(h(\hat{\underline{x}}) - h(\underline{x}_0)) \stackrel{a}{\approx} s(n^{\frac{1}{2}}(\hat{\underline{x}} - \underline{x}_0); \partial h(\underline{x}_0))$ and therefore the asymptotic distribution of $n^{\frac{1}{2}}(h(\hat{\underline{x}}) - h(\underline{x}_0))$ is the same as the distribution of $s(\underline{Y}; \partial h(\underline{x}_0))$, where $\underline{Y} \sim N(0, \Gamma)$.

Unfortunately a general theory of distribution of $s(\underline{Y}; A)$, $\underline{Y} \sim N(0, \Gamma)$, is not yet available. However, we are able to say that if functions $h(\underline{x})$ and $g(\underline{x})$ are strongly u.q.d. at $\underline{x}_0$ then

$$(5.2) \quad n^{\frac{1}{2}}(h(\hat{\underline{x}}) - h(\underline{x}_0)) \stackrel{a}{\approx} n^{\frac{1}{2}}(g(\hat{\underline{x}}) - g(\underline{x}_0))$$

if and only if $\partial h(\underline{x}_0) = \partial g(\underline{x}_0)$.

Proposition 5.2'

Let a function $g(\underline{x})$ be twice differentiable at $\underline{x}_0$, then

$$(5.3) \quad n(g(\hat{\underline{x}}) - g(\underline{x}_0)) \stackrel{a}{\approx} n^{\frac{1}{2}}\underline{y}'\nabla g(\underline{x}_0) + \frac{1}{2}\underline{y}'H\underline{y}$$

where $\underline{y} = n^{\frac{1}{2}}(\hat{\underline{x}} - \underline{x}_0)$ and $H = \frac{\partial^2 g(\underline{x}_0)}{\partial \underline{x} \partial \underline{x}'}$ is the Hessian of g at $\underline{x}_0$.

Therefore if $\nabla g(\underline{x}_0) = 0$ then the asymptotic distribution of

$n(g(\hat{\underline{x}}) - g(\underline{x}_0))$ is the same as the distribution of the quadratic form $\frac{1}{2}\underline{Y}'H\underline{Y}$, where $\underline{Y} \sim N(0, \Gamma)$.

These asymptotic results together with the min-max calculus of sections 3 and 4 provide us with the following theorems.

Theorem 5.3

Suppose that Ω is a compact topological space, $f(\underline{x}, \underline{\theta})$ is differentiable in $\underline{x}$ for all $\underline{\theta} \in \Omega$ and the gradient $\frac{\partial f(\underline{x}, \underline{\theta})}{\partial \underline{x}}$ is continuous in $\underline{x}$ and $\underline{\theta}$. Then $n^{\frac{1}{2}}(\hat{f}^* - f_0)$ has the same asymptotic distribution as the distribution of $s(\underline{Y}; A)$, where $\underline{Y} \sim N(0, \Gamma)$ and

$$(5.4) \quad A = \overline{\text{co}} \left\{ \bigcup_{\underline{\theta} \in \Omega(\underline{x}_0)} \nabla f_{\underline{\theta}}(\underline{x}_0) \right\}.$$

Therefore $n^{\frac{1}{2}}(\hat{f}^* - f_0)$ is asymptotically normal if and only if the gradients $\nabla f_{\underline{\theta}}(\underline{x}_0)$ are identical for all $\underline{\theta} \in \Omega(\underline{x}_0)$.

Proof

Follows from Theorem 3.1(a) and Proposition 5.1. ■

Theorem 5.4

Let Ω be compact and the regularity conditions of Theorem 4.2 hold. Then

(a) $n^{\frac{1}{2}}(\hat{\underline{\theta}} - \underline{\theta}_0)$ is asymptotically normal with zero means and covariance matrix

$$(5.5) \quad H_{\underline{\theta}\underline{\theta}}^{-1}, H_{\underline{\theta}\underline{x}}, \Gamma_{\underline{\theta}\underline{x}}, H_{\underline{\theta}\underline{\theta}}^{-1},$$

(b) $n^{\frac{1}{2}}(\hat{f}_* - f_0)$ is asymptotically normal with zero mean and variance $\underline{a}' \Gamma \underline{a}$, where $\underline{a} = \frac{\partial f(\underline{x}_0, \underline{\theta}_0)}{\partial \underline{x}}$

(c) Denoting $\underline{y} = n^{\frac{1}{2}}(\hat{\underline{x}} - \underline{x}_0)$ and $H = H_{\underline{x}\underline{x}} - H_{\underline{\theta}\underline{x}}' H_{\underline{\theta}\underline{\theta}}^{-1} H_{\underline{\theta}\underline{x}}$,

we have

$$(5.6) \quad n(\hat{f}_* - f_0) \stackrel{d}{=} n^{\frac{1}{2}} \underline{a}' \underline{Y} + \frac{1}{2} \underline{Y}' \underline{H} \underline{Y}.$$

In particular, if $\underline{a} = \frac{\partial f(\underline{x}_0, \underline{\theta}_0)}{\partial \underline{x}} = 0$ then the asymptotic distribution

of $n(\hat{f}_* - f_0)$ is the same as the distribution of the quadratic form $\frac{1}{2} \underline{Y}' \underline{H} \underline{Y}$, where $\underline{Y} \sim N(0, \Gamma)$.

Proof

Theorem 4.2(i) and Theorem 5.2 imply (a). (b) follows from (4.3) and Theorem 5.2. (c) follows from Theorem 4.2(ii) and Proposition 5.2.

If we have the constraint case (4.14) and the regularity conditions of Theorem 4.4 hold then we correct the results of Theorem 5.4 by sub-

stituting $\hat{\underline{\mu}} = \begin{pmatrix} \hat{\underline{\theta}} \\ \hat{\underline{\lambda}} \end{pmatrix}$ and $\underline{\mu}_0 = \begin{pmatrix} \underline{\theta}_0 \\ \underline{\lambda}_0 \end{pmatrix}$ in place of $\hat{\underline{\theta}}$ and $\underline{\theta}_0$ respectively

setting $\underline{a} = \frac{\partial L(\underline{x}_0, \underline{\mu}_0)}{\partial \underline{x}}$ and considering all Hessians to be of the

Lagrange function $L(\underline{x}, \underline{\mu})$ with respect to $\underline{\mu}$ rather than $\underline{\theta}$.

If the gradient $\underline{a}$ in (5.6) is zero (it can be the case if, for example, $f(\underline{x}, \underline{\theta})$ attains its minimum at $\underline{x} = \underline{x}_0$ for fixed $\underline{\theta} = \underline{\theta}_0$) then $n(\hat{f}_* - f_0)$

has the asymptotic distribution of the quadratic form $\frac{1}{2} \underline{Y}' \underline{H} \underline{Y}$, where

$\underline{Y} \sim N(0, \Gamma)$. On the other hand if $\underline{a} \neq 0$ then the linear part in

the right-hand side of (5.6) will dominate as $n \rightarrow \infty$ and as a consequence

$n(\hat{f}_* - f_0)$ does not converge. In any case $n^{\frac{1}{2}}(\hat{f}_* - f_0)$ converges

in distribution to normal with variance $\underline{a}' \Gamma \underline{a}$, which is nonzero if

$\underline{a} \neq 0$ (and Γ is positive definite). However, a problem may arise

when $\underline{a}$ is close to zero but not zero. Then the convergence may be slow

and the normal distribution may be a poor approximation. In this case

the distribution of the quadratic form

$$(5.7) \quad n f_0 + n^{\frac{1}{2}} \underline{a}' \underline{Y} + \frac{1}{2} \underline{Y}' \underline{H} \underline{Y}$$

may give a reasonable approximation to the distribution of $n\hat{f}_*$.

The distributions of quadratic forms are well discussed. The following result will be used in the next section (e.g., Rao and Mitra (1971) p.171).

Proposition 5.3

Let $\underline{Y} \sim N(0, \Gamma)$ and Γ be positive definite. Then the quadratic form (5.7) has the noncentral chi-square distribution $\chi^2(v, \delta)$ if and only if

- (i) $(\frac{1}{2}H)\Gamma(\frac{1}{2}H) = \frac{1}{2}H$
- (ii) There exists such vector $\underline{z}$ that $\underline{a} = H\underline{z}$
- (iii) $\underline{a}'\Gamma\underline{a} = \delta f_0$

in which case $v = \frac{1}{2}\text{tr}H\Gamma$ and $\delta = \frac{1}{2}\underline{a}'\Gamma H\underline{a}$.

The idea of using the noncentral chi-square distribution may be given a precise formulation if, following Wald (1943), we allow $\underline{x}_0$ to vary in such a manner that $\underline{x}_0$ converges at the rate $n^{-\frac{1}{2}}$ to a point $\underline{x}^*$ at

which $\frac{\partial f}{\partial \underline{x}}$ is zero (see also Stroud (1972)).

Theorem 5.5

Let $\underline{x}_0$ converge to $\underline{x}^*$ and :

- (a) The regularity conditions of Theorem 4.2 hold at $\underline{x}^*$ and $\underline{\theta}^* = \underline{\theta}(\underline{x}^*)$..

$$(b) \frac{\partial f(\underline{x}^*, \underline{\theta}^*)}{\partial \underline{x}} = 0$$

- (c) $\frac{1}{2}n(\underline{x}_0 - \underline{x}^*)'H(\underline{x}_0 - \underline{x}^*)$ converges to a number δ , where

$$H = H_{xx} - H_{\theta x}' H_{\theta\theta}^{-1} H_{\theta x}, \text{ and all Hessians are taken at } (\underline{x}^*, \underline{\theta}^*)$$

Then $n(\hat{f}_* - f_*)$ has the asymptotic distribution of $\chi^2(v, \delta)$,

where $f_* = f_*(\underline{x}^*)$ and $v = \frac{1}{2}\text{tr}H\Gamma$, if and only if

$$(5.8) \quad (\frac{1}{2}H)\Gamma(\frac{1}{2}H) = \frac{1}{2}H$$

Proof

By Theorem 4.2 and (b) we have

$$(5.9) \quad n(\hat{f}_* - f_*) \stackrel{d}{=} \frac{1}{2} \underline{Z}' H \underline{Z}$$

$$\text{where } \underline{Z} = n^{\frac{1}{2}}(\hat{\underline{x}} - \underline{x}^*) = n^{\frac{1}{2}}(\hat{\underline{x}} - \underline{x}_0) + n^{\frac{1}{2}}(\underline{x}_0 - \underline{x}^*)$$

It is known that the distribution of $\frac{1}{2} \underline{Z}' H \underline{Z}$, where $\underline{Z} \sim N(\underline{\beta}, \Gamma)$

is $\chi^2(v, \delta)$ if and only if the condition (5.8) holds. Noting that

$$\delta = \frac{1}{2} \underline{\beta}' H \underline{\beta} \quad \text{and using (c) we complete the proof.}$$

Unfortunately the applicability of Theorem 5.5 is questionable because usually vector $\underline{x}_0$ represents a fixed quantity in the population as, for instance, in covariance structure analysis $\underline{x}_0 = \text{vecs}(\Sigma)$. We

consider this problem once more in the following sections. To improve the appearance of the normal approximation to the distribution of

$n^{\frac{1}{2}}(\hat{f}_* - f_0)$ we estimate its bias in the following theorem

Theorem 5.6

Let $\hat{\underline{x}}$ be an unbiased estimate of $\underline{x}_0$. Then the bias of

$n^{\frac{1}{2}}(\hat{f}_* - f_0)$ is given by

$$(5.10) \quad \frac{1}{2} n^{-\frac{1}{2}} \text{tr } H \Gamma + o(n^{-\frac{1}{2}})$$

where H is as in Theorem 5.4(c).

Proof

From (5.6) we have

$$(5.11) \quad E\{n^{\frac{1}{2}}(\hat{f}_* - f_0)\} = E\{\underline{a}' \underline{y}\} + \frac{1}{2} n^{-\frac{1}{2}} E\{\underline{y}' H \underline{y}\} + o(n^{-\frac{1}{2}})$$

But $E\{\underline{a}' \underline{y}\} = 0$ because $\hat{\underline{x}}$ is an unbiased estimate of $\underline{x}_0$. Substituting

$$(5.12) \quad n^{-\frac{1}{2}} E\{\underline{y}' H \underline{y}\} = n^{-\frac{1}{2}} E\{\text{tr} H \underline{y} \underline{y}'\} = n^{-\frac{1}{2}} \text{tr} H \Gamma + o(n^{-\frac{1}{2}})$$

into (5.11) we complete the proof.

We remark that the unbiasedness of $\hat{\underline{x}}$ can be replaced by the condition $E\{\underline{a}' \underline{y}\} = o(n^{-\frac{1}{2}})$.

If the second order derivatives of $\underline{\theta}(\underline{x})$ are available then following the same line of proof as in Theorem 5.6 it can be shown that the bias of $n^{\frac{1}{2}}(\hat{\underline{\theta}} - \underline{\theta}_0)$ is given (in vector form) by

$$(5.13) \quad \frac{1}{2} n^{-\frac{1}{2}} \sum_{i,j=1}^m \Gamma_{ij} \frac{\partial^2 \underline{\theta}(\underline{x}_0)}{\partial x_j \partial x_i} + o(n^{-\frac{1}{2}})$$

where Γ_{ij} is the typical element of Γ .

Theoretically the required derivatives $\frac{\partial^2 \underline{\theta}}{\partial x_j \partial x_i}$ can be calculated in accordance with Theorem 4.3, although in practice it may be an awful task.

6. GENERALIZED LEAST SQUARES ESTIMATES

We apply now the general theory of the previous section to the specific case of the so-called generalized least squares (GLS) function

$$(6.1) \quad f(\underline{x}, \underline{\theta}) = (\underline{x} - \underline{\sigma}(\underline{\theta}))' V (\underline{x} - \underline{\sigma}(\underline{\theta}))$$

where V is a fixed $m \times m$ positive definite weight matrix and

$\underline{\sigma}(\underline{\theta}) = (\sigma_1(\underline{\theta}), \dots, \sigma_m(\underline{\theta}))'$ is a vector-valued function of the parameter vector $\underline{\theta} \in \Omega$. We suppose that r equality constraints

$\underline{g}(\underline{x}, \underline{\theta}) = (g_1(\underline{x}, \underline{\theta}), \dots, g_r(\underline{x}, \underline{\theta}))' = 0$ are imposed and all functions

involved are twice continuously differentiable. As in the previous section

it is assumed that $\hat{\underline{x}}$ represents an estimate of $\underline{x}_0$ and $\hat{f}_*, \hat{\underline{\theta}} = \underline{\theta}(\hat{\underline{x}})$

$\hat{\underline{\lambda}} = \underline{\lambda}(\hat{\underline{x}})$ are the corresponding estimates of $f_0 = f_*(\underline{x}_0)$, $\underline{\theta}_0$, $\underline{\lambda}_0$

respectively.

Let us calculate the required derivatives of $f(\underline{x}, \underline{\theta})$ at a point $(\underline{x}_0, \underline{\theta}_0)$

(cf., Browne (1974)).

$$(6.2) \quad \frac{\partial f}{\partial \underline{x}} = 2V(\underline{x}_0 - \underline{\sigma}_0) \quad , \quad \text{where } \underline{\sigma}_0 = \underline{\sigma}(\underline{\theta}_0)$$

$$(6.3) \quad \frac{\partial^2 f}{\partial \underline{x} \partial \underline{x}'} = 2V$$

$$(6.4) \quad \frac{\partial f}{\partial \underline{\theta}} = 2\Delta'V(\underline{\sigma}_0 - \underline{x}_0) \quad , \quad \text{where } \Delta = \frac{\partial \underline{\sigma}(\underline{\theta}_0)}{\partial \underline{\theta}'} \quad \text{is of order } m \times q$$

$$(6.5) \quad \frac{\partial^2 f}{\partial \underline{\theta} \partial \underline{\theta}'} = 2(\Delta'V\Delta + Q) \quad , \quad \text{where } Q = [(\underline{\sigma}_0 - \underline{x}_0)'V \otimes I_q] \frac{\partial \Delta}{\partial \underline{\theta}}$$

and $\frac{\partial \Delta}{\partial \underline{\theta}} = \left[\frac{\partial \Delta_{ij}}{\partial \underline{\theta}} \right]$ is of order $mq \times q$ (see Nel (1980), (5.2.1) and (5.2.5)).

$$(6.6) \quad \frac{\partial^2 f}{\partial \underline{\theta} \partial \underline{x}} = -2\Delta'V$$

All derivations will be much simpler if we assume that there exists

$\underline{\theta}_0 \in \Omega$, $\underline{g}(\underline{x}_0, \underline{\theta}_0) = 0$, such that $\underline{x}_0 = \underline{\sigma}(\underline{\theta}_0)$ (the null hypothesis).

Therefore we consider this case first. Then under the null hypothesis:

(a) $\underline{\theta}_0$ is a minimizer of f for $\underline{x} = \underline{x}_0$ and $f_0 = f_*(\underline{x}_0) = 0$.

(b) The vector of Lagrange multipliers $\underline{\lambda}_0$ is zero.

(c) The gradients $\frac{\partial f}{\partial \underline{x}}$ and $\frac{\partial f}{\partial \underline{\theta}}$ at $(\underline{x}_0, \underline{\theta}_0)$ are zero.

(d) The matrix Q of (6.5) is zero.

(e) Furthermore, even if the matrix V depends on $\underline{x}$ it does not change the derivatives (6.2) - (6.6) at $(\underline{x}_0, \underline{\theta}_0)$.

(f) The parameters $\underline{\theta}$ is c-identified at $\underline{\theta}_0$ given $\underline{x}_0$ if (the usual identification condition):

$$\underline{\sigma}(\underline{\theta}^*) = \underline{x}_0 \quad \text{implies} \quad \underline{\theta}^* = \underline{\theta}_0.$$

(g) The level condition at $\underline{x}_0$ means:

There does not exist such a sequence $\{\underline{\theta}_n\} \rightarrow \infty$, $\underline{\theta}_n \in \Omega$, such that $\underline{\sigma}(\underline{\theta}_n) \rightarrow \underline{x}_0$

Now by Theorem 5.1 we have that if $\underline{\theta}$ is identified (in the usual sense) at $\underline{\theta}_0$ given $\underline{x}_0$ and the parameter space Ω is compact then $\hat{\underline{\theta}}$ is a consistent estimate of $\underline{\theta}_0$. Note that the level condition (in the form (g)) usually holds and therefore the assumption about the compactness of Ω is unnecessary in most applications.

If in addition $\underline{\theta}_0$ is an interior point of Ω and the matrix

$$\begin{bmatrix} C_1 & C_2 \\ C_2' & C_3 \end{bmatrix} = \begin{bmatrix} \Delta' V \Delta & G_{\theta} \\ G_{\theta}' & 0 \end{bmatrix}^{-1}$$

where $G_{\theta} = \frac{\partial g'}{\partial \underline{\theta}}$, exists then by Theorem 5.4:

(i) $n^{\frac{1}{2}} \begin{bmatrix} \hat{\underline{\theta}} - \underline{\theta}_0 \\ \hat{\underline{\lambda}} \end{bmatrix}$ is asymptotically normal with zero means and the

following covariance matrix

$$(6.7) \quad \begin{bmatrix} C_1 & C_2 \\ C_2' & C_3 \end{bmatrix} \begin{bmatrix} \Delta' V \\ G_{x'} \end{bmatrix} \Gamma[V \Delta; G_{x'}'] \begin{bmatrix} C_1 & C_2 \\ C_2' & C_3 \end{bmatrix}$$

where $G_{x'} = \frac{\partial g}{\partial \underline{x}'}$.

(ii) The asymptotic distribution of $\hat{n}f_*$ is the same as the distribution of the quadratic form $\frac{1}{2}Y'HY$, where $Y \sim N(0, \Gamma)$ and

$$(6.8) \quad \frac{1}{2}H = V - \left[\frac{\Delta'V}{G_{x'}} \right]' \left[\begin{array}{c|c} C_1 & C_2 \\ \hline C_2' & C_3 \end{array} \right] \left[\frac{\Delta'V}{G_{x'}} \right]$$

In particular if $V = \Gamma^{-1}$ (if V depends on $\underline{x}$ it means that the value of V at $\underline{x}_0$ is Γ^{-1}) and $G_{x'} = 0$ (i.e., the constraints g_i are independent of $\underline{x}$) it can be verified (see Propositions 4.2 and 5.3) that the covariance matrix (6.7) takes the form

$$(6.9) \quad \left[\begin{array}{c|c} C_1 & 0 \\ \hline 0 & -C_3 \end{array} \right]$$

and $\frac{1}{2}Y'HY$ (and hence asymptotically $\hat{n}f_*$) has the chi-square distribution with $\nu = m - q + r$ degrees of freedom. Special cases of these results have been obtained by Browne (1974, 1982) and by Lee and Bentler (1980).

Now we consider what happens under an alternative hypothesis, i.e. $\underline{x}_0 \neq \underline{\sigma}(\underline{\theta}_0)$ where $\underline{\theta}_0$ is a minimizer (under the equality constraints) of f for fixed $\underline{x} = \underline{x}_0$. Theorem 5.4 can be applied and it follows that, under suitable regularity conditions,

$n^{\frac{1}{2}} \begin{pmatrix} \hat{\theta} - \underline{\theta}_0 \\ \hat{\lambda} - \underline{\lambda}_0 \end{pmatrix}$ is asymptotically normal with zero means and covariance

matrix $H_{\mu\mu}^{-1}, H_{\mu x}, H_{\mu x'}^{-1}, H_{\mu\mu}^{-1}$, where the matrix $H_{\mu\mu}$ is given by (4.21)

with

$$(6.10) \quad H_{\theta\theta} = 2 \left(\Delta'V\Delta + Q + \sum_{i=1}^r \lambda_{0i} \frac{\partial^2 g_i}{\partial \theta \partial \theta'} \right)$$

and

$$(6.11) \quad H_{\mu x}^{\mu x} = 2[-V\Delta + \sum_{i=1}^r \lambda_{0i} \frac{\partial^2 g_i}{\partial \theta^i \partial x} ; G_x']$$

Note that here $\lambda_0 = (\lambda_{01}, \dots, \lambda_{0r})'$ is not necessarily zero. If the matrix V depends on x then the corresponding derivatives also have to be taken into account.

The test statistic $n\hat{f}_*$ does not converge any more. In fact we have that $n^{1/2}(\hat{f}_* - f_0)$ is asymptotically normal with zero mean and variance

$\underline{a}'\Gamma\underline{a}$, where

$$(6.12) \quad \underline{a} = 2 \left[V(\underline{x}_0 - \underline{\sigma}_0) + \sum_{i=1}^r \lambda_{0i} \frac{\partial g_i(\underline{x}_0, \underline{\theta}_0)}{\partial \underline{x}} \right]$$

The bias of $n^{1/2}(\hat{f}_* - f_0)$ can be estimated in accordance with Theorem 5.6.

However, as has been mentioned in the previous section, the problem may arise if $\underline{x}_0$ is close to a point $\underline{x}^*$ such that $\underline{x}^* = \underline{\sigma}(\underline{\theta}^*)$ for some

$\underline{\theta}^* \in \Omega$ and $g(\underline{x}^*, \underline{\theta}^*) = 0$. Then for $V = \Gamma^{-1}$ and $G_x = 0$ the non-

central chi-square distribution $\chi^2(\nu, \delta)$ may be a reasonable approximation to the distribution of $n\hat{f}_*$ due to Theorem 5.5 (cf., Wald (1943)), where $\nu = m - q + r$ and $\delta = \frac{1}{2}n(\underline{x}_0 - \underline{x}^*)H(\underline{x}_0 - \underline{x}^*)$. Note that here we must verify the regularity conditions at the point $(\underline{x}^*, \underline{\theta}^*)$.

There are even more reasons to use the noncentral chi-square distribution if the quadratic form (5.7) has the noncentral chi-square distribution. In the following theorem we show that this is indeed so (in the unconstrained case) if $V = \Gamma^{-1}$ and the matrix Q defined in (6.5) is

zero. Note that Q is zero not only if $\underline{x}_0 = \underline{\sigma}_0$ but also if

$\frac{\partial \Delta}{\partial \underline{\theta}} = 0$, i.e., the second order derivatives of $\underline{\sigma}(\underline{\theta})$ are zero at $\underline{\theta}_0$.

This is clearly the case if $\underline{\sigma}(\underline{\theta})$ is linear in $\underline{\theta}$.

Theorem 6.1

Let $V = \Gamma^{-1}$ and $Q = 0$ then the quadratic form (5.7) has a $\chi^2(v, \delta)$ distribution with $v = m - q$ and $\delta = n(\underline{x}_0 - \underline{\sigma}_0)' V (\underline{x}_0 - \underline{\sigma}_0) = nf_0$.

Proof

We only have to verify the conditions (i) - (iii) of Proposition 5.3.

Condition (i) can be checked by direct calculation as in the case $\underline{x}_0 = \underline{\sigma}_0$ above.

Now

$$(6.13) \quad \frac{1}{2} H = \Gamma^{-1} - \Gamma^{-1} \Delta (\Delta' \Gamma^{-1} \Delta)^{-1} \Delta' \Gamma^{-1} = F (F' \Gamma F)^{-1} F'$$

where F is an $m \times (m - q)$ matrix of rank $m - q$ such that $\Delta' F = 0$ (see the matrix identity (4.31)). From (6.4) we have

$$(6.14) \quad \frac{1}{2} \frac{\partial f}{\partial \underline{\theta}} = \Delta' \Gamma^{-1} (\underline{\sigma}_0 - \underline{x}_0) = 0$$

and hence $\Delta' \underline{a} = 0$ (we recall that $\underline{a} = 2\Gamma^{-1}(\underline{x}_0 - \underline{\sigma}_0)$). Therefore $\underline{a}$ belongs to the space spanned by column vectors of F and this together with (6.13) implies (ii).

$$(6.15) \quad \underline{a}' \Gamma \underline{a} = 4(\underline{x}_0 - \underline{\sigma}_0)' \Gamma^{-1} (\underline{x}_0 - \underline{\sigma}_0) = 4f_0$$

which implies (iii). Finally

$$(6.16) \quad \begin{aligned} \delta &= \frac{1}{2} n \underline{a}' \Gamma H \underline{a} = n(\underline{x}_0 - \underline{\sigma}_0)' \Gamma^{-1} (\Gamma^{-1} - \Gamma^{-1} \Delta (\Delta' \Gamma^{-1} \Delta)^{-1} \Delta' \Gamma^{-1}) \Gamma^{-1} (x_n - \sigma_n) \\ &= n(\underline{x}_0 - \underline{\sigma}_0)' \Gamma^{-1} (\underline{x}_0 - \underline{\sigma}_0) = nf_0 \end{aligned}$$

In summary we would recommend the use of the noncentral chi-square distribution for the test statistic nf_* if $V = \Gamma^{-1}$ (or in the case when V depends on $\underline{x}$ the value of V at $\underline{x}_0$ is Γ^{-1}) and $\underline{x}_0$ is close to $\underline{\sigma}_0 = \underline{\sigma}(\underline{\theta}_0)$. Especially the $\chi^2(v, \delta)$, $\delta = nf_0$, distribution may be reasonable if $\underline{\sigma}(\underline{\theta})$ is linear in $\underline{\theta}$ (compare with the linear least squares estimation theory, e.g., Rao (1973) p.190).

7. GENERALIZED LEAST SQUARES COVARIANCE STRUCTURES ANALYSIS AND FACTOR ANALYSIS

Now we consider the results of the previous section in context of covariance structures analysis. In this respect the factor analysis model (1.1) will be an interesting and important example. Therefore we discuss it in detail and in particular we show how the problem of nonidentifiability of parameters can be treated in this case.

We start with the $p \times p$ sample covariance matrix S that is an unbiased estimate of the population covariance matrix Σ . The column vector formed from elements of S , taken columnwise will be denoted by $\text{vec}(S)$, i.e., $\text{vec}(S) = (s_{11}, s_{21}, s_{31}, \dots, s_{12}, s_{22}, s_{32}, \dots, s_{pp})'$, and

the column vector formed from the elements above and including the diagonal of a symmetric matrix S , taken columnwise, will be denoted by $\text{vecs}(S)$, i.e., $\text{vecs}(S) = (s_{11}, s_{12}, s_{22}, s_{13}, s_{23}, s_{33}, \dots, s_{pp})'$.

As the $p \times p$ matrix S is symmetric, the $p(p+1)/2 \times 1$ vector $\underline{s} = \text{vecs}(S)$ may be expressed in terms of the $p^2 \times 1$ vector $\underline{c} = \text{vec}(S)$:

$$(7.1) \quad \underline{s} = K_p' \underline{c}$$

where the matrix K_p is of order $p^2 \times p(p+1)/2$. (For a detailed discussion of the matrix K_p see Browne (1974) or Nel (1980).)

It follows from the multivariate central limit theorem that $n^{1/2}(\underline{s} - \underline{\sigma})$, where $\underline{\sigma} = \text{vecs}(\Sigma)$, is asymptotically normal with zero means and a

$p(p+1)/2 \times p(p+1)/2$ covariance matrix Γ , assuming the existence of the fourth order moments about the means. In particular, if the sample is drawn from a normally distributed population then (Browne (1974))

$$(7.2) \quad \Gamma = 2K_p'(\Sigma \otimes \Sigma)K_p.$$

Now the results of sections 5 and 6 can be applied if we set $\hat{x} = \underline{s}$ and $\underline{x}_0 = \underline{\sigma}$. The GLS function

$$(7.3) \quad f(\underline{s}, \underline{\theta}) = (\underline{s} - \underline{\sigma}(\underline{\theta}))' V (\underline{s} - \underline{\sigma}(\underline{\theta}))$$

takes the following simple form

$$(7.4) \quad f(\underline{s}, \underline{\theta}) = \frac{1}{2} \text{tr} [(S - \Sigma(\underline{\theta}))B]^2$$

if the $p(p+1)/2 \times p(p+1)/2$ positive definite matrix V is of the form

$$(7.5) \quad V = \frac{1}{2} K_p^{-1} (B \otimes B) K_p^{-1}$$

and $\underline{\sigma}(\underline{\theta}) = \text{vecs}(\Sigma(\underline{\theta}))$, where $\Sigma(\underline{\theta})$ is a symmetric matrix-valued function of $\underline{\theta}$ (Browne (1974)). Here K_p^{-1} is a left inverse of K_p

$$(7.6) \quad K_p^{-1} = (K_p' K_p)^{-1} K_p'$$

and hence the vector $\underline{c} = \text{vec}(S)$ is given by

$$(7.7) \quad \underline{c} = K_p^{-1} \underline{s}.$$

By the results of the previous section the asymptotic distribution of $n^{1/2}(\hat{\underline{\theta}} - \underline{\theta}_0)$, under the suitable regularity assumptions, is normal with zero means and the covariance matrix (unconstrained case)

$$(7.8) \quad (\Delta' V \Delta + Q)^{-1} \Delta' V \Gamma V \Delta (\Delta' V \Delta + Q)^{-1}$$

where $\Delta = \frac{\partial \underline{\Sigma}(\underline{\theta}_0)}{\partial \underline{\theta}}$ is of order $p(p+1)/2 \times q$ and the $q \times q$ matrix Q is as in (6.5). In particular, under the null hypothesis $\Sigma = \Sigma(\underline{\theta}_0)$ and if the matrices Γ and V are given by (7.2) and (7.5) respectively, then the covariance matrix (7.8) takes the following form (Browne (1974))

$$(7.9) \quad 2[\bar{\Delta}'(B \otimes B)\bar{\Delta}]^{-1}\bar{\Delta}'[(B \Sigma B) \otimes (B \Sigma B)]\bar{\Delta}[\bar{\Delta}'(B \otimes B)\bar{\Delta}]^{-1}$$

where $\bar{\Delta} = K_p^{-1}\Delta = \frac{\partial \text{vec}(\Sigma(\underline{\theta}_0))}{\partial \underline{\theta}}$ is of order $p^2 \times q$.

Under the null hypothesis the asymptotic distribution of the test statistic nf_* is a distribution of the quadratic form $\underline{Y}'HY$, where $\underline{Y} \sim N(0, \Gamma)$ and the matrix H is given by (6.8). It is known that the quadratic form $\underline{Y}'HY$ is distributed as

$$(7.10) \quad \underline{Y}'HY \sim \sum_{i=1}^k \lambda_i Z_i^2$$

where $Z_1, \dots, Z_k$ denote independent $\chi^2(1)$ variates and λ_i are the nonzero eigenvalues of the matrix $H\Gamma$. (e.g., Rao (1973)).

In particular, if Γ is given by (7.2) (normally distributed population) and V is given by (7.5) with $B = \Sigma^{-1}$ (or, e.g., $B = S^{-1}$)

then all eigenvalues λ_i equal one and $k = m - q + r$ (i.e.,

$(H\Gamma)^2 = H\Gamma$) and hence we have the chi-square distribution with

$v = m - q + r$ degrees of freedom (Browne (1974), Lee and Bentler (1980)).

Note that the normality assumption (7.2) is essential here (cf., Kent (1982), White (1982)). For a more detailed discussion of the nonnormal case see Browne (1982).

Under the alternative hypothesis (and suitable assumptions) we showed that the noncentral chi-square distribution may be a reasonable approximation to the distribution of nf_* if Σ is close to $\Sigma(\underline{\theta}_0)$ for some $\underline{\theta}_0 \in \Omega$.

Let us now consider the factor analysis model (1.1). Unfortunately the parameter Λ is never identified, since any orthogonal rotation of Λ provides the same value of $\Sigma(\theta)$. Consequently Λ is not estimable and Theorem 5.4 is not applicable (the matrix $\Delta'V\Delta$ is singular).

However, Ψ is usually identified (see Anderson and Rubin (1956) section 5) and we are willing to find its asymptotic distribution. At the first step we eliminate Λ by minimizing the discrepancy function (7.4) with respect to Λ . Therefore we define the function $h(S)$ of a $p \times p$ symmetric matrix S as follows

$$(7.11) \quad h(S) = \min_{\Lambda} \text{tr}[(S - \Lambda\Lambda')B]^2$$

where Λ is of order $p \times k$. Note that $h(S)$ can be considered as the function of $p(p+1)/2$ independent variables given by vector $\underline{s} = \text{vecs}(S)$.

Lemma 7.1

Let B be nonsingular and $S_0 = AA'$ where A is a $p \times k$ matrix of rank k . Then $h(\underline{s})$ is twice continuously differentiable at $\underline{s}_0 = \text{vecs}(S_0)$, $\nabla h(\underline{s}_0) = 0$ and

$$(7.12) \quad \frac{\partial^2 h(\underline{s}_0)}{\partial \underline{s} \partial \underline{s}'} = K_p^{-1}(Z \otimes Z)K_p^{-1}$$

where $Z = E(E'B^{-1}E)^{-1}E'$ and E is a $p \times (p-k)$

matrix of rank $p-k$ such that $E'A = 0$.

Proof

First we observe that

$$(7.13) \quad \text{tr}[(S - \Lambda\Lambda')B]^2 = (\underline{c} - \text{vec}(\Lambda\Lambda'))'(B \otimes B)(\underline{c} - \text{vec}(\Lambda\Lambda'))$$

where $\underline{c} = \text{vec}(S)$. It is clear that for $S = S_0$ the minimum in (7.11) is attained at $\Lambda = AT$, where T can be any $k \times k$ orthogonal matrix.

Therefore, as mentioned earlier, Λ is not identified. However, Λ can be identified by imposing exactly $k(k-1)/2$ equality constraints (e.g., Lawley and Maxwell (1971)). Then, treating $\underline{c}$ in (7.13) as a vector of independent variables, we obtain (see (4.33)):

$$(7.14) \quad \frac{\partial h(\underline{s}_0)}{\partial \underline{c} \partial \underline{c}'} = B \otimes B - (B \otimes B) \bar{\Delta} W [W' \bar{\Delta}' (B \otimes B) \bar{\Delta} W]^{-1} W' \bar{\Delta}' (B \otimes B)$$

where $\bar{\Delta} = \left. \frac{\partial \text{vec}(\Lambda \Lambda')}{\partial \text{vec}'(\Lambda)} \right|_{\Lambda=A}$ is of order $p^2 \times pk$ and W (as

in (4.33)) is a matrix orthogonal to G_0 of full column rank $pk - k(k-1)/2$.

A particular choice of W is not important. What is important is that $\bar{\Delta} W$ now has full column rank $pk - k(k-1)/2$. Then using some matrix calculus (see Nel (1980)) we have

$$(7.15) \quad \frac{\partial^2 h(\underline{s}_0)}{\partial \underline{s} \partial \underline{s}'} = V - V \Delta (\Delta' V \Delta)^{-1} \Delta' V$$

where $V = K_p' (B \otimes B) K_p^{-1}$ and $\Delta = K_p' \bar{\Delta} W$ is of full column rank

$pk - k(k-1)/2$. Now, using the matrix identity (4.31), expression (7.15) can be written as follows

$$(7.16) \quad \frac{\partial^2 h(\underline{s}_0)}{\partial \underline{s} \partial \underline{s}'} = F (F' V^{-1} F)^{-1} F'$$

where F is of full column rank

$$p(p+1)/2 - [pk - k(k-1)/2] = (p-k)(p-k+1)/2$$

such that $F' \Delta = 0$. The matrix V^{-1} is given by (Browne (1974))

$$(7.17) \quad V^{-1} = K_p' (B^{-1} \otimes B^{-1}) K_p$$

Let us show that we can take F as follows

$$(7.18) \quad F = K_p' (E \otimes E) K_{p-k}$$

We have (see Nel (1980))

$$(7.19) \quad \bar{\Delta} = (A \otimes I_p)' + (I_p \otimes A)I_{(k,p)}$$

and therefore for F given by (7.18)

$$(7.20) \quad F' \Delta = K_{p-k}' (E' \otimes E') M_p \bar{\Delta} W = K_{p-k}' (E' \otimes E') \bar{\Delta} W = \\ K_{p-k}' [(E' A) \otimes E' + (E' \otimes E' A) I_{(k,p)}] W = 0$$

where $M_p = K_p' K_p^{-1}$ and $M_p \bar{\Delta}' = \bar{\Delta}'$ (see Browne (1974)).

Finally substituting F , in the form (7.18), into (7.16) and using (7.17) it is not difficult to obtain (7.12). ■

Let the column vector formed from the diagonal elements of a matrix S be denoted by $\text{diag}(S)$. Then

$$(7.21) \quad \text{diag}(S) = D_p' \text{vec}(S)$$

where D_p is the $p^2 \times p$ matrix (see Browne (1974), Nel (1980)).

We define the function f of $\underline{s} = \text{vecs}(S)$ and $\underline{\varphi} = \text{diag}(\Psi)$ as follows

$$(7.22) \quad f(\underline{s}, \underline{\varphi}) = h(S - \Psi)$$

where the function h is defined by (7.11). If $\Sigma = AA' + \Psi$ then the required derivatives of f at $\underline{x} = \underline{\sigma} = \text{vecs}(\Sigma)$ and $\underline{\varphi} = \text{diag}(\Psi)$ can be written as follows (see Lemma 7.1):

$$(7.23) \quad \frac{\partial^2 f}{\partial \underline{s} \partial \underline{s}'} = K_p' (Z \otimes Z) K_p^{-1}$$

$$(7.24) \quad \frac{\partial^2 f}{\partial \underline{\varphi} \partial \underline{s}'} = - D_p' (Z \otimes Z) K_p^{-1}$$

$$(7.25) \quad \frac{\partial^2 f}{\partial \underline{\varphi} \partial \underline{\varphi}} = - D_p' (Z \otimes Z) D_p$$

Note that (Browne (1974)):

$$(7.26) \quad D_p'(V \otimes W)D_p = V * W$$

where $V * W$ represents the term by term product of V and W (Hadamard product). Therefore $D_p'(Z \otimes Z)D_p = Z * Z$ and

$$D_p'D_p = D_p'(I_p \otimes I_p)D_p = I_p * I_p = I_p.$$

The Hadamard square $Z * Z$ will be denoted by $Z^{(2)}$.

Theorem 7.1

Let $\hat{\varphi}$ be the GLS sample estimate of $\varphi = \text{diag}(\Psi)$ under the null hypothesis $\Sigma = AA' + \Psi$ and suppose:

(a) Ψ is identified, i.e., $\Sigma = \bar{A}'\bar{A} + \bar{\Psi}$ implies $\Psi = \bar{\Psi}$.

(b) The $p \times p$ matrix $Z^{(2)}$ is nonsingular, where Z is specified in Lemma 7.1.

Then $n^{1/2}(\hat{\varphi} - \varphi)$ is asymptotically normal with zero means and covariance matrix

$$(7.27) \quad (Z^{(2)})^{-1}D_p'(Z \otimes Z)D_p(Z^{(2)})^{-1}$$

with under the normality assumption (7.2) becomes

$$(7.28) \quad 2(Z^{(2)})^{-1}(Z \Psi Z)(Z^{(2)})^{-1}$$

Proof

Formula (7.27) follows by Theorem 5.4(a) from (7.24), (7.25) and (7.26). Under the normality assumption (7.28) follows from (7.27) by (7.26) and

$$Z\Sigma = ZAA' + Z\Psi = Z\Psi, \text{ since } ZA = E(E'B^{-1}E)^{-1}E'A = 0. \quad \blacksquare$$

In particular for $B = \Sigma^{-1}$ we have

$$Z \Sigma Z = E(E' \Sigma E)^{-1} E' \Sigma E (E' \Sigma E)^{-1} E' = Z$$

and expression (7.28) becomes $2(Z^{(2)})^{-1}$ (Jöreskog and Goldberger (1972))

8. MINIMUM TRACE FACTOR ANALYSIS

In this section we discuss the asymptotic properties of the so-called minimum trace factor analysis (MTFA). The example of MTFA is of interest because the nonidentifiability leads here to the non-differentiability and therefore the techniques developed in section 3 are of a particular relevance in this case.

Let Σ be the population covariance matrix and $\underline{w} = (w_1, \dots, w_p)'$ a chosen vector of weights. Consider the following problem

$$(8.1) \quad \max_{\Psi} \underline{w}' \Psi \underline{w}$$

subject to Ψ diagonal and $\Sigma - \Psi$ Gramian. We shall refer to it as the MTFA problem with weight vector $\underline{w}$. In the case of unit weights the maximization problem (8.1) is equivalent to minimization of trace of $\Sigma - \Psi$, therefore the name. The MTFA has applications to the reliability theory (see Bentler (1972), Jackson and Agunwamba (1977), Bentler and Woodward (1980)). The coefficient $\rho(\underline{w})$, that is a lower bound to reliability, is defined as follows

$$(8.2) \quad \rho(\underline{w}) = 1 - \frac{\underline{w}' \Psi^* \underline{w}}{\underline{w}' \Sigma \underline{w}}$$

where Ψ^* is the solution of (8.1).

We also consider the maximization of $\rho(\underline{w})$ with respect to $\underline{w}$ and define

$$(8.3) \quad \rho^* = \max_{\underline{w} \neq 0} \rho(\underline{w})$$

We shall refer to (8.3) as the weighted minimum trace factor analysis (WMTFA) problem (Shapiro [in press] (b)). Substituting Σ by the sample covariance matrix S we obtain the sample estimates $\hat{\rho}(\underline{w})$ and $\hat{\rho}^*$ of $\rho(\underline{w})$ and ρ^* respectively. The asymptotic properties of $\hat{\rho}(\underline{w})$ and $\hat{\rho}^*$ will be of interest to us. In the following lemma we give the max-representation of $\hat{\rho}(\underline{w})$ and ρ^* (Bentler and Woodward (1980)), Ten Berge, Snijders and Zegers (1981)).

Lemma 8.1

$$(8.4) \quad \hat{\rho}(\underline{w}) = \max_{Z \in \Omega} f(S, Z, \underline{w})$$

$$(8.5) \quad \hat{\rho}^* = \max_{(Z, \underline{w}) \in \Omega \times S^{p-1}} f(S, Z, \underline{w})$$

where Ω is the set of Gramian matrices with unit diagonal elements,

$$S^{p-1} = \{\underline{w} : \underline{w}'\underline{w} = 1\} \quad \text{and}$$

$$(8.6) \quad f(S, Z, \underline{w}) = 1 - \frac{\text{tr } S W Z W}{\underline{w}' S \underline{w}}$$

with such diagonal matrix W that $\text{diag}(W) = \underline{w}$.

We denote $\Omega(S)$ and $\Omega^*(S)$ the subsets of Ω and $\Omega \times S^{p-1}$ respectively where the maximum in (8.4) and (8.5) is attained. The gradient ∇f of f with respect to $\underline{s} = \text{vec}(S)$ at $\underline{s} = \underline{\sigma} = \text{vec}(\Sigma)$, Z and $\underline{w}$ is

$$(8.7) \quad \nabla f = \alpha \text{vec}(\underline{w} \underline{w}') - \beta \text{vec}(W Z W)$$

where $\alpha = (\text{tr } \Sigma W Z W) (\underline{w}' \underline{\Sigma} \underline{w})^{-2}$ and $\beta = (\underline{w}' \underline{\Sigma} \underline{w})^{-1}$. Then Theorem 5.3 can be applied and we obtain the following result.

Theorem 8.1

$n^{\frac{1}{2}}(\hat{\rho}(\underline{w}) - \rho(\underline{w}))$ and $n^{\frac{1}{2}}(\hat{\rho}^* - \rho^*)$ have the asymptotic distribution as the distribution of $s(\underline{Y}; A)$ and $s(\underline{Y}; A^*)$ respectively, where $\underline{Y} \sim N(0, \Gamma)$, Γ is the $p^2 \times p^2$ asymptotic covariance matrix of $n^{\frac{1}{2}}(\underline{s} - \underline{\sigma})$ and the sets

A and A^* are given by

$$(8.8) \quad A = \text{co} \left\{ \bigcup_{Z \in \Omega(\Sigma)} \nabla f \right\}$$

$$(8.9) \quad A^* = \text{co} \left\{ \bigcup_{(Z, \underline{w}) \in \Omega^*(\Sigma)} \nabla f \right\}$$

and ∇f is given by (8.7).

Corollary 8.1

$n^{\frac{1}{2}} \hat{\rho}(\underline{w}_0)$ is asymptotically equal to $n^{\frac{1}{2}} \hat{\rho}^*$ if and only if the maximum in (8.3) (or equivalently in (8.5) for $S = \Sigma$) is attained at unique (up to multiplication by a scalar) vector $\underline{w} = \underline{w}_0$.

Corollary 8.2

$n^{\frac{1}{2}}(\hat{\rho}(\underline{w}) - \rho(\underline{w}))$ is asymptotically normal if and only if $\Omega(\Sigma)$ contains only one matrix Z_0 . In the last case the asymptotic variance is $\underline{a}' \Gamma \underline{a}$, where

$$(8.10) \quad \underline{a}' = \alpha \text{vec}(\underline{w} \underline{w}') - \beta \text{vec}(W Z_0 W)$$

In particular under the normality assumption (7.2) the asymptotic variance becomes (Shapiro (1982))

$$(8.11) \quad 2\alpha^2 (\underline{w}' \Sigma \underline{w})^2 - 4\alpha\beta \underline{w}' \Sigma W Z_0 W \Sigma \underline{w} + 2\beta^2 \text{tr}(W Z_0 W \Sigma)^2$$

The uniqueness of solution in (8.4) and (8.5) has been studied in detail by Shapiro (1982), Shapiro [in press] (b).

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